

Ken Yamashiro

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Education

University of Southern California – B.S. Mechanical Engineering, GPA 3.71 Expected Graduation May 2027

Experience

USC Viterbi School of Engineering — Unmanned Aerial Systems Project December 2025 – Present

Undergraduate Assistant and Lead Student Engineer Los Angeles, CA

- Assist Dr. Penkova in characterizing flight dynamics of organic, bio-inspired aerodynamic designs on low-Reynolds-number autonomous VTOL drones in hybrid water/air applications
- Develop electrical systems, program flight dynamics, and essential systems for VTOL flight

Niagara Bottling May 2026 – Present

Packaging Research and Development Intern Diamond Bar, CA

- Apply AI and machine learning algorithms to ANSYS FEA tools to simulate and validate lightweighting designs, reducing simulation turnaround time by over 1000 hours for the engineering team
- Integrate simulation outputs into design iterations to support production and environmental improvements

NASA Columbia Scientific Balloon Facility May 2025 – July 2025

Intern/Undergraduate Assistant Palestine, TX

- Supported the final production and payload integration of the Payload for Ultrahigh Energy Observations (PUEO) research project, a NASA Pioneers project successfully launched in December 2025
- Performed surface preparation, resolved tolerance conformities, and flight preparation of 300+ components
- Effectively collaborated with over 50 researchers from 11 universities across the U.S., UK, and Taiwan

UH Department of Physics and Astronomy, High Energy Physics Group (HEPG) September 2023 – August 2025

Lab Tech II and Researcher Honolulu, HI

- Utilized Siemens SolidEdge, FEA, and drafting to design, fabricate, and test 8+ physics research projects focused on particle physics, radio frequency (RF), and space environments.
- Used CNC, CAM, manual milling, lathes, and additive manufacturing to fabricate 700+ flight hardware parts
- Prepared technical materials and assisted in proposal presentations to NASA, JPL, and various universities

Extracurricular Activities

USC Formula SAE Racing | Powertrain Engineer August 2025 - Present

- Designed, manufactured, and tested powertrain and cooling subsystem components for the SCR26 race car (600cc inline-4 Yamaha R6) using Siemens NX, ANSYS, and MATLAB
- Selected from 60+ members to present in the design competition at the 2026 Michigan FSAE competition
- Supported assembly, physical testing, and analysis of the SCR26 car, helping place 7th out of 110 teams

Project Experience (More listed on GitHub Website)

Payload for Ultrahigh Energy Observations (PUEO) NASA

- Antarctic detector for Neutrinos and ultrahigh energy particles, part of the NASA Pioneers Program
- Fabricated 400+ flight components from various materials, programmed CNC toolpaths, and performed FEA
- Drafted part documentation and 100+ reference drawings used during assembly

Askaryan Calorimeter Experiment (ACE) Fermilab/UH HEPG

- Designed and fabricated an RF testing assembly, resolving part tolerances, assembly, and manufacturability
- Performed FEA on RF assembly to validate waveform compliance and structural stability

Skills and Certifications

Skills: SolidWorks, Siemens NX, Siemens SolidEdge, ANSYS, FEA analysis, Manual/CNC Machining, 3D printing, MATLAB

Experience in: Additive/Subtractive manufacturing, 6061, 7075 Al, Brass 360, CF tubes/honeycomb, 100/101 Cu, more

SOLIDWORKS CAD Design Associate Certification, Certificate ID: C-SY4HKFETCM